

Presentation Title:

Mental Models in Artificial Intelligence: Understanding Developer Thinking

Slide 1: Introduction — Why This Topic Matters

- AI is not just code—it's a reflection of the mind behind it.
- Developers build AI systems based on how they *think* intelligence works.
- Understanding their mental models is essential to understanding AI behavior.

Speaker Cue: "Behind every AI decision is a developer's mind shaping it. Let's explore how."

Slide 2: What Are Mental Models? (Psychology Perspective)

- Mental models = internal cognitive representations of external reality
- Help with decision-making, prediction, and learning
- Used unconsciously in everyday life

"We don't see reality—we interpret it through our mental models."

Slide 3: Mental Models in the Mind of a Developer

- Developers predict how code will behave before running it
- They visualize logic, behavior, and outcomes in advance
- These mental simulations guide their design choices

Visual: Brain with flowchart overlay

Slide 4: Developers Don't Code—They Think in Abstractions

- AI is built using abstractions: layers, parameters, logic blocks
- These are mentally modeled before implementation
- The *mental architecture* often shapes the actual codebase

Thought-Provoker: "Is the AI just a shadow of the developer's mental architecture?"

Slide 5: Flawed Mental Models → Flawed AI

- If a developer's understanding is limited or biased, the AI inherits that flaw
- Ex: Misunderstanding human behavior → biased recommendation engines
- AI failures often trace back to human cognitive assumptions

Slide 6: Feedback Loops in AI Design

- Developers create models → Models generate output → Developers interpret and tweak
- Their mental model evolves *with* the AI
- This loop can strengthen or worsen biases

Note: "Mental models are not static—they evolve with experience and system feedback."

Slide 7: Cognitive Biases in Developer Thinking

- Common biases that affect AI development:
 - **Confirmation bias** (favoring data that supports their hypothesis)
 - **Overconfidence** in model generalization
 - **Anchoring** to initial results
- These biases can embed themselves in the AI system



Slide 8: Misalignment Between Developer & User Mental Models

- Developers understand systems in-depth—users don't
- Users may misinterpret AI behavior (e.g., "Why did this recommendation appear?")
- UX failures are often *mental model mismatches*

Visual: Venn diagram showing Developer vs. User vs. AI mental models



Slide 9: Explainable AI (XAI) – A Bridge Between Minds

- XAI aims to align AI behavior with *human* expectations
- Helps users form accurate mental models of AI
- Transparency = trust

Thought-Provoker: "Can an AI explain itself in a way that makes sense to a human brain?"



Slide 10: Tools to Build Better Developer Mental Models

- Visual debugging tools
- Interactive modeling platforms (e.g., TensorBoard, Streamlit)
- Code review and pair programming to cross-check models

Speaker Cue: "Better tools = better thinking = better AI."



Slide 11: Shared Mental Models in Team Environments

- AI is rarely built alone—team cohesion requires aligned mental models

- Misunderstandings lead to system inconsistency
- Psychological safety improves model sharing and feedback



Slide 12: Case Study — Self-Driving Cars

- Developers simulate how humans perceive roads
- Errors in modeling human perception → dangerous outcomes
- Tesla & Uber incidents show gaps between mental simulation and real-world complexity



Slide 13: The Future — Can AI Understand *Our* Mental Models?

- Cognitive modeling: AI predicting how humans think
- AI tutors, AI therapists, human-AI collaboration
- Next frontier: Co-evolving mental models of human and machine



Slide 14: Thought-Provoking Questions

- Are we designing AI—or are we projecting ourselves into it?
- Can we train developers in *metacognition* to improve AI?
- What if an AI could critique *your* mental model?



Slide 15: Conclusion

- AI is shaped by the minds of its creators
- Understanding developer thinking is key to understanding AI behavior
- Psychology and computer science must work together to build ethical, aligned, intelligent systems

Quote to End With:

“We shape our tools, and thereafter, our tools shape us.” – Marshall McLuhan